



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

AIOT

06-05-2023

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	C,D	B	A,B,D	B,D	C,D	B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	D	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	720.00	18.85	6.00	1.33	240.00	2.00	10.00	60.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C,D	A	B,C,D	B,D	B	B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	D	C	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.00	3.00	15.00	23.00	4.00	10.00	8.00	75.75

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	A,B	A,B,C,D	A,B	A,C	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	C	D	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	-25.00	1.50	5040.00	3.00	52.00	4.50	12.50	-26.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C,D)

Current potentiometer wire $I = 2A$

$$V_{AB} = 4V$$

$$\text{Potential gradient (x)} = \frac{V_{AB}}{L} = \frac{1}{30} V/cm$$

$$(A) V = \frac{2}{4} \times 1 = \frac{1}{2} V$$

$$\text{Balancing length} = \frac{V}{x} = 15cm$$

Similarly for (B) $V = 1.5 V$, (C) $V = 1 V$, (D)

$$V = 2 V$$

2. Ans (B)

$$r = \frac{mV}{qB}$$

$$\lambda = \frac{h}{mV} = \frac{h}{rqB}$$

$$M = \frac{q}{2m} \cdot mV r = \frac{q}{2m} \times r^2 qB$$

$$M = 0.8 \times 10^{-11} Am^2$$

3. **Ans (A,B,D)**

Gravitation field outside the cylindrical planet

$$E = \frac{G \cdot 2\pi\rho R^2}{r}$$

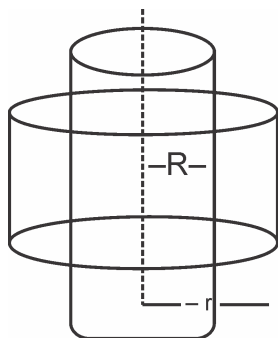
$$mE = \frac{mv^2}{r} = \frac{G \cdot 2\pi\rho R^2 m}{r}$$

$$KE = \frac{mv^2}{2} = G \cdot R\pi\rho R^2$$

$$\text{again } m\omega^2 r = \frac{G \cdot 2\pi\rho R^2 m}{r}$$

$$T^2 \propto r^2$$

$$T \propto r$$



4. **Ans (B,D)**

Smaller the least count greater the precision.

$$\text{Mean absolute error } |\Delta x| = \frac{\sum |x - x_i|}{N}$$

5. **Ans (C,D)**

29 division of main scale coincides with 30 divisions of vernier scale Hence one division of vernier scale

$$= \frac{30 - 29}{30} \text{ of main scale}$$

$$= \frac{1}{30} \times 0.5^\circ$$

$$= \frac{1}{30} \times 0.5 \times 60 \text{ min}$$

$$= 1 \text{ min}$$

Position (1) $20^\circ 20'$

Position (2) $30^\circ 16'$

Position (3) $32^\circ 39'$

Position (4) $33^\circ 41'$

$$D_V = 33^\circ 41' - 20^\circ 20' = 13^\circ 21'$$

$$D_y = 32^\circ 39' - 20^\circ 20' = 12^\circ 19'$$

6. **Ans (B,C)**

Just after release acceleration of O must be zero.

$$\text{So } a = \frac{\alpha R}{2}$$

$$\frac{mgR}{2} = \frac{1}{2} \frac{mR^2}{2} \times \omega^2$$

$$\omega = \sqrt{\frac{2g}{R}}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (A)**

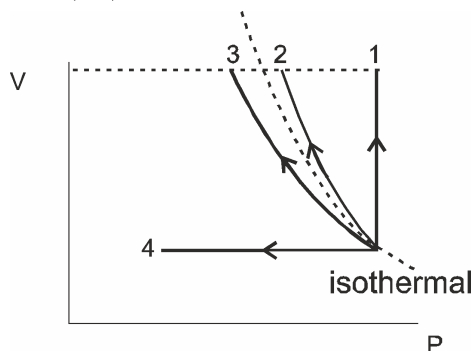
$$a = -w^2 y$$

$$V = \pm w \sqrt{A^2 - y^2}$$

$V_{\text{particle}} = -(\text{slope}) \times \text{velocity of wave}$

Kinetic energy and P.E. both are maximum at mean position.

8. **Ans (D)**



No work is done during process 4 for because volume is not changing.

Work done is maximum in process 1 one because area of PV graph is maximum.

For process 1 and 2 temperature will increase because PV is increasing.

9. Ans (D)

Diagram showing a parallel plate capacitor with plates separated by distance $d + Vt$. The capacitance is given by $C_A = \frac{\epsilon_0 S}{d + Vt}$.

Diagram showing a parallel plate capacitor with plates separated by distance $d - Vt$. The capacitance is given by $C_B = \frac{\epsilon_0 S}{d - Vt}$.

$$\frac{C_A}{C_B} = \frac{d - Vt}{d + Vt}$$

$$\frac{Q_A}{Q_B} = \frac{C_A}{C_B} = \frac{d - Vt}{d + Vt}$$

$$Q_A + Q_B = 2Q$$

$$Q_A = \frac{(d - Vt)Q}{d}$$

$$i = \frac{dQ_B}{dt} = -\frac{dQ_A}{dt} = \frac{QV}{d}$$

$$\frac{U_A}{U_B} = \frac{C_A}{C_B} = \frac{d - Vt}{d + Vt}$$

10. Ans (A)

(A) Apply momentum conservation in the direction of initial velocity of balls.

$$mv + mv = mv' + mv' + 2mv'$$

$$v' = \frac{2v}{4} \Rightarrow \frac{v}{2}$$

(B) For speed apply energy conservation

$$\frac{1}{2}mv^2 + \frac{1}{2}mv^2 = \frac{1}{2}2mv'^2 + 2\left(\frac{1}{2}mv_b^2\right)$$

$$mv^2 = mv'^2 + mv_b^2$$

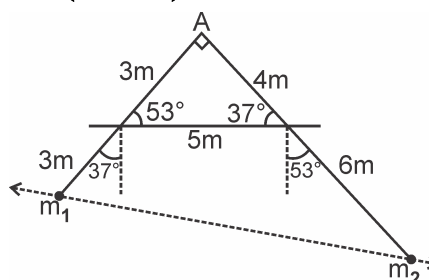
$$v^2 - \frac{v^2}{4} = v_b^2 \Rightarrow \frac{3v^2}{4} = v_b^2$$

$$\Rightarrow v_b = \frac{\sqrt{3}}{2} V$$

PART-1 : PHYSICS

SECTION-II

1. Ans (720.00)



Net torque about A = 0

$$m_1 g \sin 37^\circ \times 6 = m_2 g \sin 53^\circ \times 10$$

$$m_1 \times \frac{3}{5} \times 6 = m_2 \times \frac{4}{5} \times 10$$

$$m_2 = \frac{1.6 \times 3 \times 6}{4 \times 10}$$

$$= 0.72 \text{ kg} = 720 \text{ g}$$

2. Ans (18.85)

Projection along x and y axis will be

$$\text{SHM with } \omega = \sqrt{\frac{8}{2}} = 2 \text{ rad/s}$$

$$x = 3 \sin \omega t$$

$$y = 2 \cos \omega t$$

Path enclosed in elliptical $a = 3, b = 2$

$$\text{Area} = \pi ab = 18.85 \text{ m}^2$$

3. Ans (6.00)

For circuit-I :

$$I_1 = \frac{E}{R}$$

For circuit-II :

Current through battery is steady state &

$$I = \frac{3E}{2R}$$

$$I_2 = \frac{1}{3} I = \frac{E}{2R}$$

4. Ans (1.33)

For second surface

$$n_1 = 1.5$$

$$n_2 = 1$$

$$u = \infty$$

$$R = -\frac{20}{\pi}$$

$$\frac{1}{V} - \frac{1.5}{\infty} = \frac{1 - 1.5}{-30} \times \pi \Rightarrow V = \frac{60}{\pi} \text{ cm}$$

$$h = v\theta = \frac{60}{\pi} \times \frac{4\pi}{180} = 1.33 \text{ cm}$$

5. Ans (240.00)

$$R = \frac{u^2 \sin 2\theta}{g} = 240\text{m}$$

6. Ans (2.00)

Elongation of both springs should be always same and both particles should have same frequency of oscillation.

For 2m particle : $nkx - kx = 2ma_1$

$$a_1 = \frac{(n-1)kx}{2m}$$

For m particle : $kx = ma_2$

$$a_2 = \frac{k}{2m}(2x)$$

For same frequency :

$$\frac{(n-1)k}{2m} = \frac{k}{2m} \Rightarrow n = 2$$

7. Ans (10.00)

$$I = \int \vec{J} \cdot d\vec{a}$$

$$= \int \sigma \vec{E} \cdot d\vec{a}$$

$$I = \frac{1}{\rho} \int \vec{E} \cdot d\vec{a}$$

$$I\rho = \phi = 10 \text{ V.m}$$

8. Ans (60.00)

$$P = E_0 I_0 \sin^2 \omega t \cos \phi + E_0 I_0 \sin \omega t \cos \omega t \sin \phi$$

$$P = \frac{E_0 I_0}{2} (1 - \cos 2\omega t) \cos \phi + \frac{E_0 I_0}{2} \sin 2\omega t$$

$$\sin \phi$$

$$P = \frac{E_0 I_0}{2} \cos \phi - \frac{E_0 I_0}{2} \cos (2\omega t + \phi)$$

$$P_{\max} = \frac{E_0 I_0}{2} (\cos \phi + 1) = 3P$$

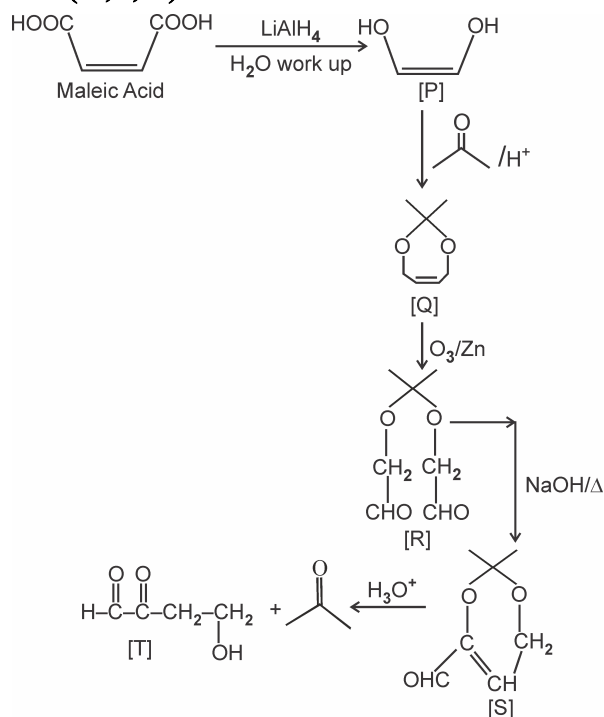
$$P_{\min} = \frac{E_0 I_0}{2} (\cos \phi - 1) = -P$$

$$\Rightarrow \cos \phi = \frac{1}{2} \Rightarrow \phi = \frac{\pi}{3}$$

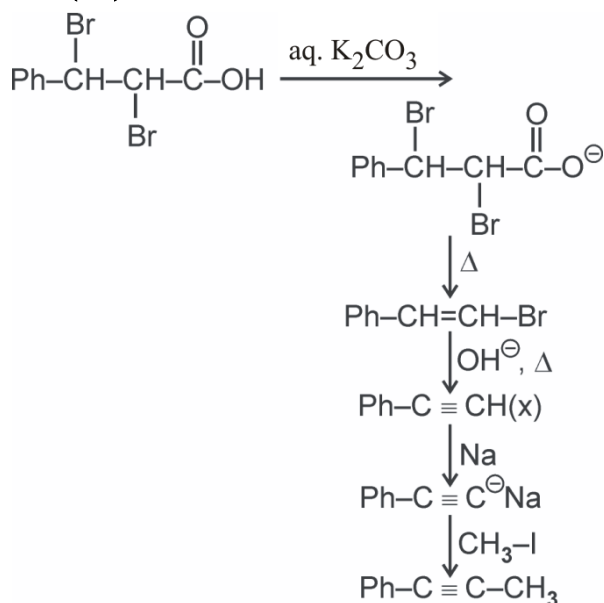
PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (B,C,D)



2. Ans (A)



3. Ans (B,C,D)

(i) belongs to 1s.

(ii) belongs to 3s.

(iii) belongs to 2s.

(iv) belongs to 3p.

4. **Ans (B,D)**

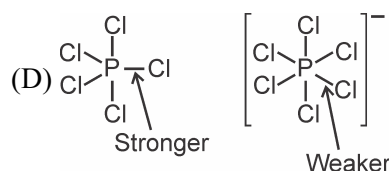
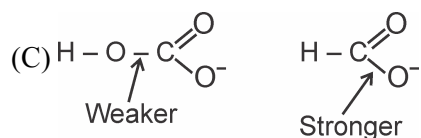
Remaining liquid will be less volatile than original.

$$\text{Also, } \frac{P_A^o}{P_B^o} = \frac{X_B Y_A}{X_A Y_B} > 1$$

5. **Ans (B)**

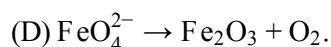
(A) Bond order diamond < Graphite.

(B) % s character in C-H bond $\triangle > \text{—}$



6. **Ans (B,C,D)**

(A) In presence of HCl, $\text{Al}(\text{OH})_3$ will be converted in AlCl_3 hence not precipitated.



PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (C)**

Theoretical

8. **Ans (D)**

K_p or K_c are not affected in any case of List-I.

For (P) effect of increased volume dominates over effect of increased moles of NO_2 .

[Think similarly for (R) and (S)]

9. **Ans (C)**

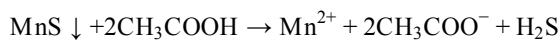
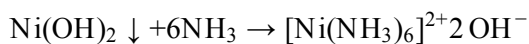
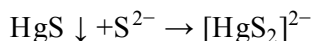
(P) H_2 shows +ve deviation in these conditions.

(Q) V_m is < 22.4 l/mol.

(R) $T > T_B$ So $Z > 1$

(S) Density $\rightarrow 0$, So $Pv = nRT$.

10. **Ans (C)**

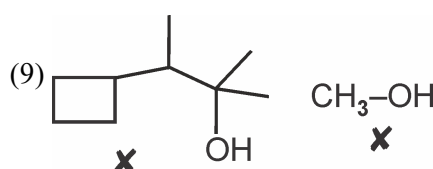
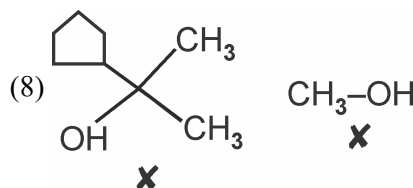
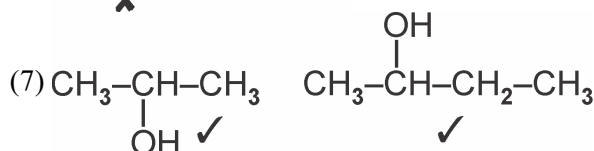
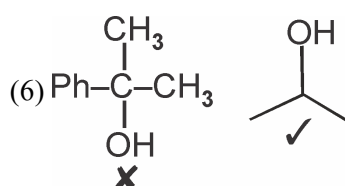
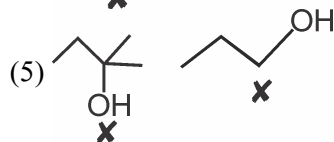
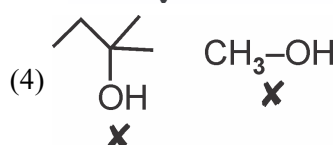
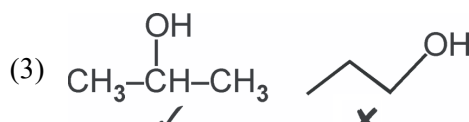
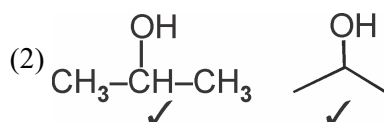
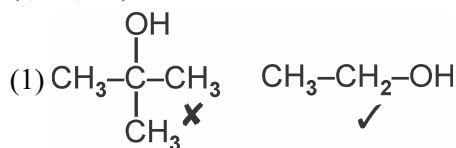


PART-2 : CHEMISTRY

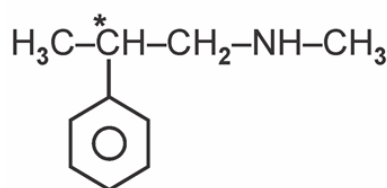
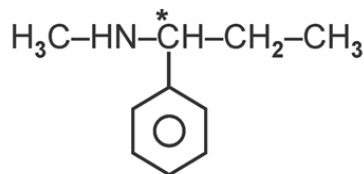
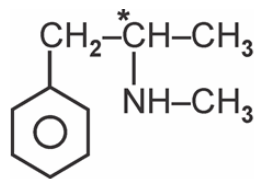
SECTION-II

1. **Ans (3.00)**

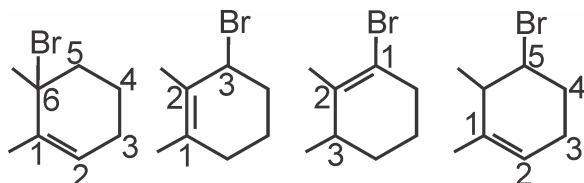
(I, III, VI)



2. Ans (3.00)



3. Ans (15.00)



$$6 + 3 + 1 + 5 = 15$$

4. Ans (23.00)

$$8 + 5 + 10 = 23$$

12 tetrahedral voids of a hexagonal prism are shared and their effective contribution is 4 in a unit cell. Remaining 8 are completely inside.

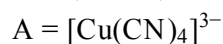
5. Ans (4.00)

$$P_1^H = pka + \log \frac{x}{a}$$

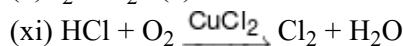
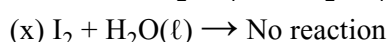
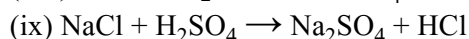
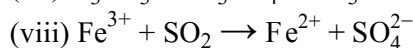
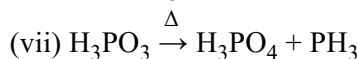
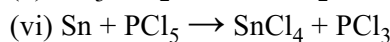
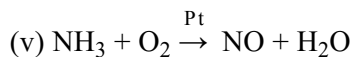
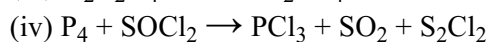
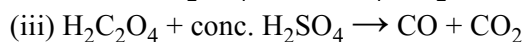
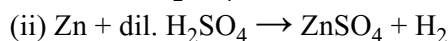
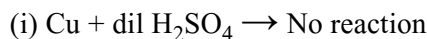
$$P_2^H = pka + \log \frac{y}{a}$$

$$P_2^H - P_1^H = 0.6$$

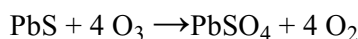
6. Ans (10.00)



7. Ans (8.00)



8. Ans (75.75)



$$\frac{303}{4} = 75.75 \text{ gram}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,C)

Plane is perpendicular to $(2\hat{i} + 3\hat{j} + 4\hat{k})$ and $(5\hat{i} + 3\hat{j} + 4\hat{k})$

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 5 & 3 & 4 \end{vmatrix} = 12\hat{j} - 9\hat{k} = 3(4\hat{j} - 3\hat{k})$$

Plane passes through point (0, 1, 2)

$$(x - 0)0 + (y - 1)4 + (z - 2)(-3) = 0$$

$$4y - 3z + 2 = 0$$

Line L is line of intersection of both planes.

$$\text{D.r. of Line} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 4 & -3 \\ 2 & 3 & 4 \end{vmatrix} = 25\hat{i} - 6\hat{j} - 8\hat{k}$$

Point on line is common point of 2 planes.

$$x = 0, \quad 3y + 4z = 5$$

$$4y - 3z = -2$$

$$25y = 7$$

$$y = \frac{7}{25}, \quad z = \frac{26}{25}$$

$$L : \frac{x-0}{25} = \frac{y-\frac{7}{25}}{6} = \frac{z-\frac{26}{25}}{-8}$$

$$\left(25\lambda, 6\lambda + \frac{7}{25}, -8\lambda + \frac{26}{25} \right)$$

2. **Ans (A,B)**

$$(A) |k \operatorname{adj}(kA)| = k^n \cdot |\operatorname{adj}(kA)|$$

$$= k^n \cdot |k^{n-1} \cdot \operatorname{adj} A|$$

$$= k^n \cdot k^{n^2-n} \cdot |A|^{n-1}$$

$$= k^{n^3-n^2} \cdot |A|^{n-1}$$

(B) Order of C is not mentioned.

$$(C) A^{-1}(AB) \Rightarrow B = 0$$

$$(D) \text{ If } |AB| \neq 0 \Rightarrow |A| \neq 0 \text{ and } |B| \neq 0$$

3. **Ans (A,B,C,D)**

$$\text{Let } Q(x) = f(x) - x^2 \begin{matrix} \swarrow 1 \\ \searrow 3 \\ \quad \alpha \end{matrix}$$

$$f(x) = (x-1)(x-3)(x-\alpha) + x^2$$

$$f(1) = 2$$

$$\alpha = 1$$

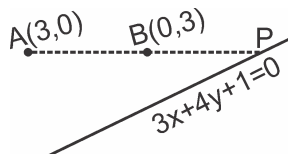
$$f(x) = (x-1)^2(x-3) + x^2$$

$$= (x^2 - 2x + 1)(x-3) + x^2$$

$$f(x) = x^3 - 4x^2 + 7x - 3$$

4. **Ans (A,B)**

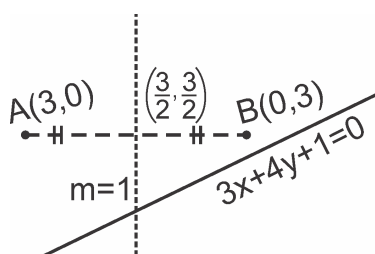
$$|PA - PB|_{\max} = AB$$



$$AB : x + y = 3, L : 3x + 4y + 1 = 0$$

$$P(13, -10)$$

$$|PA - PB|_{\min} = 0.$$



$\perp$ bisector of AB

$$y = x$$

$$P\left(-\frac{1}{7}, -\frac{1}{7}\right)$$

5. **Ans (A,C)**

$$\int_0^x 4 + |2-t| dt$$

$$= \int_0^2 (6-t) dt + \int_2^x t + 2 dt$$

$$= 6(2) - \left(\frac{4-0}{2}\right) + \left(\frac{t^2}{2} + 2t\right)_2^x$$

$$= 10 + \frac{x^2-4}{2} + 2(x-2)$$

$$\frac{x^2}{2} + 2x + 4$$

$$f(x) = \begin{cases} \frac{x^2}{2} + 2x + 4 & x > 2 \\ 5x & x \leq 2 \end{cases}$$

Continuous at $x = 2$ but not differentiable at $x = 2$.

6. **Ans (A,B,C)**

$$\text{Given } f(x+5) \geq f(x) + 5 \quad \dots(1)$$

$$f(x+1) \leq f(x) + 1 \quad \dots(2)$$

from (1) and (2), we can write

$$f(x) + 5 \leq f(x+5) \leq f(x+4) + 1$$

$$\leq f(x+3) + 2$$

$$\leq f(x+2) + 3$$

$$\leq f(x+1) + 4$$

$$\leq f(x) + 5$$

$\therefore$ equality exist everywhere

$$\therefore f(x+1) = f(x) + 1 \quad \therefore f(1) = 1$$

$$\Rightarrow f(2) = 2, f(3) = 3 \dots f(2013) = 2013.$$

$$\text{Now } g(x) = f(x) + 5 - x \Rightarrow g(2013) = 5$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B)

$$(I) \int_a^b (f(x) - 3x) dx = 2 \int_b^a x dx$$

$$\int_a^b (f(x) - 3x + 2x) dx = 0$$

$$\int_a^b (f(x) - x) dx = 0$$

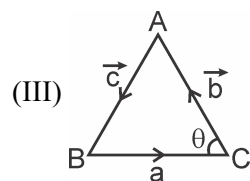
$$f(x) = x$$

$$f\left(\frac{\pi}{6}\right) = \frac{\pi}{6}$$

$$(II) \frac{1}{\pi} \times \frac{\pi^2}{\ln 3} \ln |\sec \pi x + \tan \pi x| \Big|_{7/6}^{5/6}$$

$$= \frac{\pi}{\ln 3} (\ln \sqrt{3} + \ln \sqrt{3})$$

$$= \pi$$



$$\cos \theta = \frac{4+4-12}{2 \cdot 2 \cdot 2} = -\frac{1}{2}$$

$$\theta = \frac{2\pi}{3}$$

$$(IV) I = \int_0^{\infty} \frac{dx}{(1+x^3)(1+x^2)} \quad \dots(1)$$

$$x = \frac{1}{t}$$

$$I = \int_0^{\infty} \frac{\frac{1}{t^2} dt}{\frac{(t^3+1)(t^2+1)}{t^5}}$$

$$I = \int_0^{\infty} \frac{t^3 dt}{(t^3+1)(t^2+1)} \quad \dots(2)$$

by (1) + (2)

$$2I = \int_0^{\infty} \frac{dx}{1+x^2} = \tan^{-1} x \Big|_0^{\infty} = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

8. Ans (C)

$$(I) h(x) = f(x)^2 + (f'(x))^2$$

$$h'(x) = 2f(x) \cdot f'(x) + 2f'(x) \cdot f''(x)$$

$$= 2f'(x) (f(x) + f''(x))$$

$$= -2xg(x) \cdot f'(x)^2$$

$$\text{as } h'(0^-) > 0, h'(0^+) < 0 \text{ \& } h(0) = 0$$

$$\therefore h(x) \text{ has its maxima at } x = 0$$

$$\Rightarrow k = 0 \Rightarrow k^2 + 9 = 9$$

$$(II) g(x) = f^2(x) + f(x)$$

$$g'(x) = 2f(x) \cdot f'(x) + f'(x)$$

$$= f'(x) (2f(x) + 1)$$

$$\text{given that } g'(x) \geq 0$$

$$\Rightarrow f'(x) \cdot (2f(x) + 1) \geq 0$$

$$\text{also given that } f'(x) \geq 0$$

$$\Rightarrow 2f(x) + 1 \geq 0$$

$$f(x) \geq -\frac{1}{2}$$

$$f(x)_{\min} = -\frac{1}{2}$$

$$|8(k-1)| = 12$$

$$(III) \text{ Let } G(x) = \left(\int_0^x f(t) \cdot dt \right)^2$$

$$G'(x) = 2f(x) \cdot \int_0^x f(t) dt$$

$$G'(x) = 2g(x) \text{ and } g(0) = 0 = G'(0)$$

$g(x)$ is non increasing.

$$\Rightarrow g(x) \leq 0 \forall x \in [0, \infty), g(x) \geq 0 \forall x \in (-\infty, 0)$$

$$\Rightarrow G'(x) \leq 0 \forall x \in [0, \infty), G'(x) \geq 0 \forall x \in (-\infty, 0)$$

But $G(0)$ and $G(x) \geq 0$

$$\Rightarrow G(x) = 0 \forall x \in \mathbb{R}$$

$$\int_0^x f(t) dt = 0$$

$$f(x) = 0 \quad f(1.5) + 4 = 4.$$

$$(IV) f(x) = x^2 + \int_0^x e^{-t} f(x-t) dt$$

$$f(x) = x^2 + \int_0^x e^{-(x-t)} f(t) dt$$

$$f(x) = x^2 + e^{-x} \int_0^x e^t \cdot f(t) dt$$

$$f'(x) = 2x + f(x) - e^{-x} \int_0^x e^t \cdot f(t) dt$$

$$f'(x) = 2x + f(x) - (f(x) - x^2)$$

$$f'(x) = 2x + x^2$$

$$f(x) = x^2 + \frac{x^3}{3} + c$$

$$f(0) = 0 \Rightarrow c = 0$$

$$f(x) = x^2 + \frac{x^3}{3}$$

$$f(2) = \frac{20}{3} \quad k = 20$$

9. **Ans (D)**

$$(I) \text{ Equation of chord } \frac{x \cdot 0}{25} + \frac{y \cdot 2}{36} - 1 = \frac{4}{36} - 1$$

$$y = 2$$

Put in ellipse

$$\frac{x^2}{25} + \frac{4}{36} = 1$$

$$\frac{x^2}{25} = 1 - \frac{1}{9} = \frac{8}{9}$$

$$x^2 = \frac{200}{9}$$

$$x = \pm \frac{10\sqrt{2}}{3}$$

$$\text{Length of chord} = \frac{20\sqrt{2}}{3}$$

$$(II) x^2 - 2(a - 2x)^2 = 9$$

$$-7x^2 + 8ax - 2a^2 = 9$$

$$7x^2 - 8ax + 2a^2 + 9 = 0$$

$$D = 0$$

$$64a^2 - 28(2a^2 + 9) = 0$$

$$8a^2 = 289$$

$$a^2 = \frac{63}{2}$$

$$|a| = 3\sqrt{\frac{7}{2}}$$

$$(III) y = mx - 2m - m^3$$

$$y = -2x + 4 + 8$$

$$y = -2x + 12$$

$$\sqrt{\lambda} = 2\sqrt{3}$$

$$(IV) P_1 \cdot P_2 = b^2 = 15 = k$$

$$\sqrt{k} = \sqrt{15}$$

10. **Ans (A)**

$$(I) \Delta = 0 = \begin{vmatrix} 1 & -2 & -1 \\ 3 & 3 & \sqrt{\lambda} \\ 2 & -1 & 2 \end{vmatrix} = 0$$

$$\sqrt{\lambda} = 9$$

$$\lambda = 81$$

$$(II) 3^{113} = (27)^{37} \cdot 3^2$$

$$= (26 + 1)^{37} \cdot 3^2$$

$$= 9 \left(\underbrace{{}^{37}C_0 \cdot 26^{37} + \dots}_{\text{Div. by 13}} + {}^{37}C_{37} \cdot 26^0 \right)$$

$$\text{Remainder} = 9.$$

$$(III) \text{ General term} = \frac{12!}{p!q!r!} 2^{p/2} \cdot 3^{q/3} \cdot 5^{r/6}$$

$$p + q + r = 12$$

Possible triplets

$$(0, 6, 6) \rightarrow 3$$

$$(0, 0, 12) \rightarrow 3$$

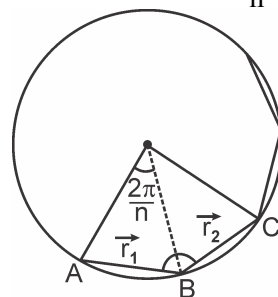
$$p \rightarrow 0, 2, 4, 6, 8, 10, 12$$

$$q \rightarrow 0, 3, 6, 9, 12$$

$$r \rightarrow 0, 6, 12$$

$$\text{Total} = 6 \text{ terms.}$$

$$(IV) \angle ABC = \pi - \frac{2\pi}{n}$$



$$\vec{r}_1 \times \vec{r}_2 = |\vec{r}_1||\vec{r}_2| \sin \frac{2\pi}{n} \hat{a}$$

$$\vec{r}_2 \times \vec{r}_3 = |\vec{r}_2||\vec{r}_3| \sin \frac{2\pi}{n} \hat{a}$$

.

.

.

$$\vec{r}_n \times \vec{r}_1 = |\vec{r}_n||\vec{r}_1| \sin \frac{2\pi}{n} \hat{a}$$

and

$$\vec{r}_1 \cdot \vec{r}_2 = -|\vec{r}_1||\vec{r}_2| \cos \frac{2\pi}{n}$$

$$\vec{r}_2 \cdot \vec{r}_3 = -|\vec{r}_2||\vec{r}_3| \cos \frac{2\pi}{n}$$

.

.

.

$$\vec{r}_n \cdot \vec{r}_1 = -|\vec{r}_n||\vec{r}_1| \cos \frac{2\pi}{n}$$

put these values in given equation, we will get

$$\text{So, } \tan \frac{2\pi}{n} = 1$$

$$\frac{2\pi}{n} = m\pi + \frac{\pi}{4}, m \in \mathbb{I}$$

$$n_{\min} = 8.$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (-25.00)**

$$\sum_{r=1}^4 \alpha_r = \sum_{r=1}^4 \beta_r = 2 - 510i$$

Let line L in parametric form be

$$z = -5 + 6i + \lambda(a + bi)$$

where slope is b/a.

$$\sum_{r=1}^5 \beta_r = -20 + 24i + k(a + bi)$$

$$\text{where } k = \sum_{r=1}^5 \lambda_r$$

$$-20 + 24i + ka + kbi = 2 - 510i$$

$$ka - 20 = 2, \quad kb + 24 = -510$$

$$ka = 22, \quad kb = -534$$

$$\text{Slope} = \frac{b}{a} = \frac{-534}{22} = -24.27 = \lambda$$

$$\Rightarrow [\lambda] = [-24.27] = -25.00$$

2. **Ans (1.50)**

He can reach coaching by either 4 or 6 steps.

EENN EEENNW or EENNNNS
(He should not take EENN as he reaches coaching in 4 steps)

$$\frac{4!}{2!2!} \quad 2 \left(\frac{6!}{3!2!} - \frac{4!}{2!2!} \times 2 \right)$$

EN
WE

$$P = \frac{6}{4^4} + \frac{96}{4^6} = \frac{1}{4^4}(12) = \frac{3}{64}$$

$$32P = \frac{3}{2} = 1.5$$

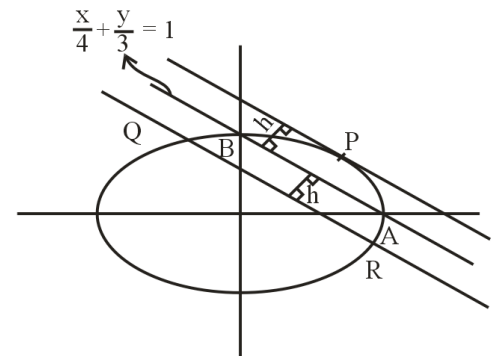
3. **Ans (5040.00)**

Total - (Row 1 is empty) - (Row 2 is empty) - (Row 3 is empty)

$$= ({}^{11}C_8 - {}^9C_8 - {}^8C_8 - {}^9C_8) \cdot \frac{8!}{4!2!}$$

4. **Ans (3.00)**

Let perpendicular distance of P from the line be h



$$\frac{1}{2} \times h \times 5 = 6(\sqrt{2} - 1)$$

$$(\text{as } \Delta PAB = 6(\sqrt{2} - 1))$$

$$h = \frac{12(\sqrt{2} - 1)}{5}$$

Now distance of tangent parallel to AB i.e.

$$4y + 3x = 12\sqrt{2},$$

$$\text{from line AB is } \frac{12(\sqrt{2} - 1)}{5}. \text{ There are just}$$

three such points.

5. **Ans (52.00)**

$$R_1 \rightarrow R_1 - R_2$$

$$R_2 \rightarrow R_2 - R_3$$

$$\Delta = \begin{vmatrix} 2 & -2 & 0 \\ 0 & 2 & -2 \\ [x] & [y] & [z] + 2 \end{vmatrix}$$

$$\Delta = 4([x] + [y] + [z] + 2)$$

$$\Delta_{\max} = 4(4 + 3 + 4 + 2)$$

$$= 52$$

6. **Ans (4.50)**

$$2\sec^3 x - \cos \frac{16}{9}x \cdot \sec x - 2\tan^2 x \cdot \sec x$$

$$= \sin \frac{x}{3} + \cos \frac{x}{3} \tan x$$

$$2\sec x - \cos \frac{16}{9}x \cdot \sec x = \frac{\sin \frac{x}{3} \cos x + \cos \frac{x}{3} \sin x}{\cos x}$$

$$2 - \cos \frac{16}{9}x = \sin \frac{4x}{3}$$

$$\sin \frac{4x}{3} + \cos \frac{16x}{9} = 2$$

$$\Rightarrow \sin \frac{4x}{3} = 1 \quad \cos \frac{16x}{9} = 1$$

$$\frac{4x}{3} = (4n+1)\frac{\pi}{2} \quad \frac{16x}{9} = 2m\pi$$

$$x = (4n+1)\frac{3\pi}{8} \quad x = \frac{9m\pi}{8}$$

$$x = \frac{3\pi}{8}, \frac{15\pi}{8}, \frac{27\pi}{8}, \dots \quad x = 0, \frac{9\pi}{8}, \frac{18\pi}{8}, \frac{27\pi}{8}$$

Sum of common solution

$$= \frac{27\pi}{8} + \frac{63\pi}{8} - \frac{9\pi}{8} - \frac{45\pi}{8} = \frac{36\pi}{8}$$

$$\Rightarrow k\pi = \frac{36\pi}{8}$$

$$\Rightarrow k = \frac{36}{8} = 4.50$$

7. **Ans (12.50)**

$$\text{Let } Y = y - 5, \quad X = x + 2$$

$$\frac{dY}{dX} = \frac{(X+Y)^2 - Y^2}{X^2}$$

$$\frac{dY}{dX} = \frac{X^2 + 2XY}{X^2}$$

$$\frac{dY}{dX} = 1 + 2 \cdot \frac{Y}{X}$$

$$\text{Let } \frac{Y}{X} = v$$

$$v + \frac{Xdv}{dX} = 1 + 2v$$

$$\int \frac{dv}{1+v} = \int \frac{dX}{X}$$

$$\ln(1+v) = \ln X + \ln C$$

$$1 + \frac{y-5}{x+2} = C(x+2)$$

$$x+y-3 = -\frac{1}{2}(x+2)^2$$

$$y = -\frac{1}{2} \times 25 = -12.5$$

8. **Ans (-26.00)**

$$p+q = -r \quad \dots(i)$$

$$r+s = -p \quad \dots(ii)$$

$$pq = s \quad \dots(iii)$$

$$rs = q \quad \dots(iv)$$

adding (i) & (ii)

$$\Rightarrow p+q+r+s = -(p+r)$$

$$\Rightarrow 2(p+r) + (q+s) = 0$$

$$\Rightarrow (q+s) = -2(p+r) \quad \dots(v)$$

Also r is a root of $x^2 + px + q = 0$

$$r^2 + pr + q = 0 \quad \dots(vi)$$

Similarly

$$p^2 + pr + s = 0 \quad \dots(vii)$$

by (vi) + (vii)

$$p^2 + r^2 + 2pr + q + s = 0$$

using (v)

$$(p+r)^2 - 2(p+r) = 0$$

$$p+r = 0(\times), (2) (\checkmark)$$

by (iii) $\times$ (iv)

$$pr = 1$$

$$p = 1, r = 1$$

$$q = -2, s = -2$$

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	C	A	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,D	B,C	B,D	A,B,C	A,D	B,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	8	7	4	2	3	4	2	1

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	B	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C	A,B,C,D	A,B,C	A,B,D	A,C,D	B,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	5	6	5	2	6	3	3	7

PART-3 : MATHEMATICS

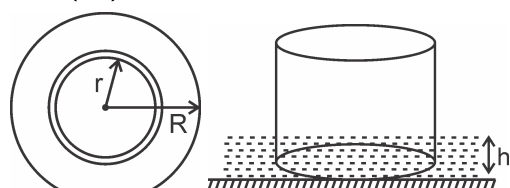
SECTION-I (i)	Q.	1	2	3	4				
	A.	C	C	D	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,B,C,D	A,B	A,C	B,C,D	A,D	B,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	5	7	9	4	7	2	4	1

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)



$$\begin{aligned}
 \int d\tau &= \int_0^R \eta (2\pi r dr) \left(\frac{r\omega - 0}{h} \right) \times r \\
 &= \frac{2\pi\eta\omega}{h} \int_0^R r^3 dr \\
 &= \frac{2\pi\eta\omega}{h} \left[\frac{R^4}{4} \right] = \frac{\pi\eta\omega R^4}{2h} \\
 \text{Power} = \tau\omega &= \frac{\pi\eta\omega^2 R^4}{2h}
 \end{aligned}$$

2. Ans (C)

$$I = I_0 + I_0 + 2\sqrt{I_0 I_0} \cos \theta$$

$$\cos \theta = 0$$

$$\theta = \frac{\pi}{2}$$

3. Ans (A)

The pitch of screw gauge, $P = 1\text{ mm}$

$$\text{least count, L.C.} = \frac{P}{N} = \frac{1}{100} = 0.01\text{ mm}$$

Diameter of the wire,

$$d = 2\text{ mm} + 56 \times \text{L.C.} - 6 \times \text{L.C.}$$

$$= 2\text{ mm} + 56 \times 0.01\text{ mm} - 6 \times 0.01\text{ mm}$$

$$d = 2.50\text{ mm} = 0.250\text{ cm}$$

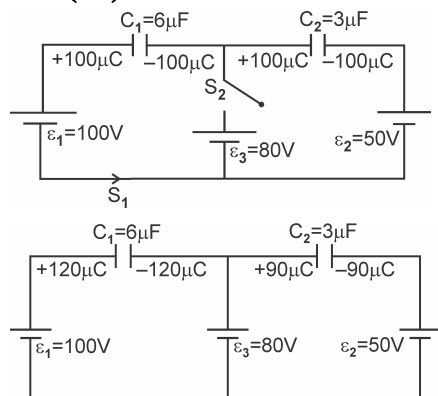
$$\ell = 8.68\text{ cm}$$

Curved surface area of the wire,

$$S = \pi d \ell$$

$$= \frac{22}{7} \times 0.250 \times 8.68 = 6.82\text{ cm}^2$$

4. Ans (B)



The charge flown through the battery ϵ_1 and ϵ_2 after closing the switch ' S_2 ',

$$\Delta Q_1 = 120 - 100 = 20 \text{ mC}$$

$$\Delta Q_2 = -90 - (-100) = 10 \text{ mC}$$

$$\text{Now, } \Delta W_b = \Delta U + \Delta H$$

$$20 \times 10^{-6} \times 100 + 10 \times 10^{-6} \times 50 - 30 \times 10^{-6} \times 80 =$$

$$\frac{1}{2} \times 6 \times 10^{-6} \times 400 + \frac{1}{2} \times 3 \times 10^{-6} \times 900$$

$$- \frac{1}{2} \times 2 \times 10^{-6} \times 2500 + \Delta H$$

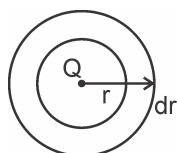
$$\Rightarrow 100 \times 10^{-6} = 50 \times 10^{-6} + \Delta H$$

$$\Rightarrow \Delta H = 50 \times 10^{-6} = 50 \mu\text{J}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (B,D)



$$\int dq = \int \frac{\alpha}{r} 4\pi r^2 dr$$

$$q = 4\pi\alpha \left[\frac{r^2}{2} \right]_R^r$$

$$= 2\pi\alpha[r^2 - R^2]$$

$$\frac{k(Q+q)}{r^2} = E$$

$$E = \frac{k(Q + 2\pi\alpha(r^2 - R^2))}{r^2}$$

$$E = \frac{kQ}{r^2} + 2\pi\alpha k - \frac{2\pi\alpha k R^2}{r^2}$$

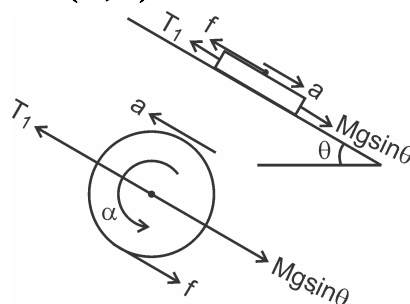
For E to be independent of r.

$$\frac{kQ}{r^2} = \frac{2\pi\alpha k R^2}{r^2}$$

$$Q = 2\pi\alpha R^2$$

$$\text{For this } E = 2\pi\alpha k = 2\pi\alpha \times \frac{1}{4\pi\epsilon_0} = \frac{\alpha}{2\epsilon_0}$$

6. Ans (B,C)



$$Mg\sin\theta - T_1 - f = Ma \dots (i)$$

$$T_1 - mg\sin\theta - f = ma \dots (ii)$$

$$fR = \frac{mR}{2} \alpha$$

$$f = \frac{mR}{2} \left(\frac{2a}{R} \right)$$

$$f = ma \dots (iii)$$

$$(i) + (ii) \Rightarrow Mg\sin\theta - mg\sin\theta - 2f = (M+m)a$$

$$(M-m)g\sin\theta = (M+m)a + 2ma$$

$$a = (M-m)g\sin\theta / (M+3m)$$

7. Ans (B,D)

For max separation velocity of both should be same and that is at $t = 1$ sec. and $t = 1.5$ sec.

Separation at $t = 1$ sec. is 2m and that at $t = 1.5$ sec. is 2.5m

8. Ans (A,B,C)

$$V_0 = I_0 R = 10 \times 10 = 100$$

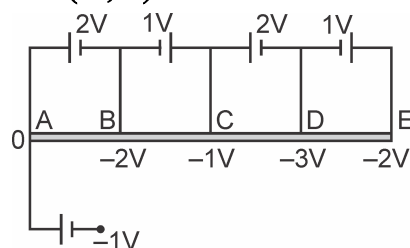
$$I = I_0 e^{-t/RC}$$

$$C = \frac{t}{R \ln \left(\frac{I_0}{I} \right)} = \frac{1}{10 \ln 2}$$

$$\text{heat produced} = \frac{1}{2} \left(\frac{1}{10 \ln 2} \right) (100)^2$$

$$= \frac{500}{\ln 2} \text{ joules}$$

9. Ans (A,D)

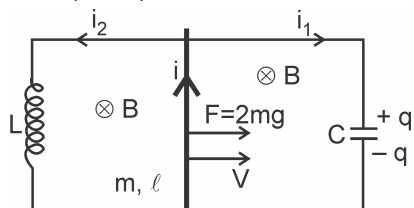


There will be null point

(1) between A and B.

(2) At C.

10. Ans (B,D)



$$x = 0$$

$$v = 0$$

$$t = 0$$

$$e = Bv\ell$$

$$q = C\varepsilon = CBv\ell$$

$$i_1 = \frac{dq}{dt} = CB\ell \frac{dv}{dt} \quad \dots\dots(i)$$

$$\frac{L di_2}{dt} = Bv\ell$$

$$\int di_2 = \left(\frac{B\ell}{L} \right) \int v dt$$

$$\int_0^{i_2} di_2 = \frac{B\ell}{L} \int_0^x dx$$

$$i_2 = \left(\frac{B\ell}{L} x \right) \quad \dots\dots(ii)$$

$$m \frac{dv}{dt} = F - Bi\ell$$

$$\frac{mdv}{dt} = F - B\ell(i_1 + i_2)$$

$$\frac{mdv}{dt} = F - B\ell i_1 - B\ell i_2$$

$$m \frac{dv}{dt} = 2mg - B\ell \left(CB\ell \frac{dv}{dt} \right) - B\ell \left(\frac{B\ell}{L} x \right)$$

$$(m + CB^2\ell^2) \frac{d^2x}{dt^2} = \frac{B^2\ell^2}{L} \left(x - \frac{2mgL}{B^2\ell^2} \right)$$

$$\frac{d^2x}{dt^2} = - \frac{B^2\ell^2}{2mL} \left(x - \frac{2mgL}{B^2\ell^2} \right) [\because CB^2\ell^2 = m]$$

$\therefore$ angular frequency of SHM executed by the rod

$$\omega = \frac{B\ell}{\sqrt{2mL}}$$

The amplitude of SHM executed by the rod is

$$A = \frac{2mgL}{B^2\ell^2}$$

The maximum horizontal displacement of the rod

$$X_{\max} = 2A = \frac{4mgL}{B^2\ell^2}$$

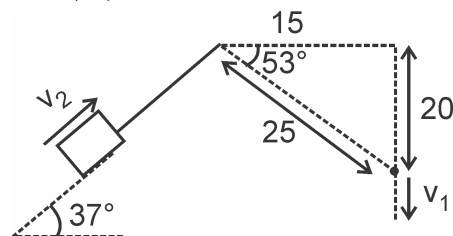
PART-1 : PHYSICS

SECTION-III

1. Ans (8)

$$F = - \frac{dU}{dr} = - \frac{1}{2} 2mb^2r = -mb^2r$$

2. Ans (7)



$$-10 \times \frac{10 \times 3}{5} + 10 \times 20 = \frac{1}{2} 1V_1^2 + \frac{1}{2} \times 1 \times V_2^2$$

$$140 = \frac{V_1^2 + V_2^2}{2}$$

$$V_1 \times \frac{4}{5} = V_2$$

$$280 = V_1^2 + \frac{16}{25} V_1^2$$

$$V_1 = \sqrt{\frac{280 \times 25}{41}}$$

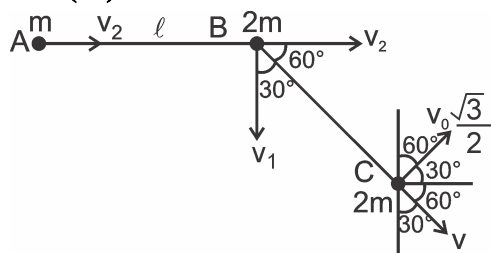
$$V_1 = \sqrt{\frac{7000}{41}}$$

3. Ans (4)

$$T = \frac{2\pi m}{qB}$$

$$T = \pi \text{ sec}$$

4. Ans (2)



Using conservation of momentum of system.

$$2mv_0 = 3mv_2 + 2m \left(\frac{3v_0}{4} + \frac{v}{2} \right)$$

$$2v_0 = 3v_2 + v + \frac{3v_0}{2}$$

$$3v_2 + v = \frac{v_0}{2} \dots\dots (i)$$

$$2mv_1 = 2m \left(\frac{\sqrt{3}v_0}{4} - \frac{v\sqrt{3}}{2} \right)$$

$$v_1 = \frac{\sqrt{3}v_0}{4} - \frac{\sqrt{3}v}{2}$$

$$\sqrt{3}v_1 = \frac{3v_0}{4} - \frac{3v}{2} \dots(ii)$$

$$\text{Also, } v_1 \cos 30^\circ + v_2 \cos 60^\circ = v$$

$$\frac{\sqrt{3}v_1}{2} + \frac{v_2}{2} = v$$

$$\sqrt{3}v_1 = 2v - v_2 \dots(iii)$$

From equation (ii) and (iii)

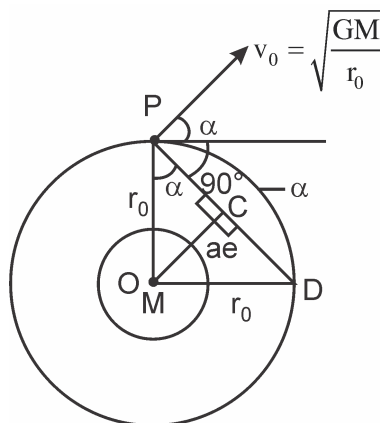
$$2v - v_2 = \frac{3v_0}{4} - \frac{3v}{2}$$

$$7v - 2v_2 = \frac{3v_0}{2} \dots(iv)$$

$$23v_2 = 2v_0$$

$$v_1 = \frac{2v_0}{23} = \frac{2 \times 23}{23} = 2 \text{ m/s}$$

5. Ans (3)



Total mechanical energy

$$\begin{aligned} \epsilon &= \frac{-GMm}{r_0} + \frac{1}{2}mv_0^2 \\ &= -\frac{GMm}{r_0} + \frac{1}{2}m\left(\sqrt{\frac{GM}{r_0}}\right)^2 = -\frac{GMm}{r_0} + \frac{GMm}{2r_0} \end{aligned}$$

$$\begin{aligned} \epsilon &= -\frac{GMm}{2r_0} \\ \frac{-GMm}{2a} &= -\frac{GMm}{2r_0} \end{aligned}$$

$\therefore$ Length of semi-major axis, $a = r_0$

Hence point 'P' must be an extreme

Point of the minor axis of the elliptical orbit

$\therefore \sin \alpha = ae$

$$e = \sin \alpha = \sin 37^\circ = 0.6$$

Hence the eccentricity of the elliptical orbit

$$e = 0.60$$

6. Ans (4)

The tension produced in each wire,

$$F = \frac{(\alpha_1 + \alpha_2) \ell \Delta \theta}{\frac{\ell}{A} \left(\frac{1}{Y_1} + \frac{1}{Y_2} \right)} = \frac{3 \times 10^{-5} \times 20 \times 10^{-5}}{1.5 \times 10^{-10}} = 40 \text{ N}$$

$$\text{Now, } \frac{n_1}{2\ell} \sqrt{\frac{F}{\mu_1}} = \frac{n_2}{2\ell} \sqrt{\frac{F}{\mu_2}}$$

$$\Rightarrow \frac{n_1}{n_2} \sqrt{\frac{\mu_1}{\mu_2}} = \sqrt{\frac{0.4}{0.1}} = 2$$

In first overtone, $n_1 = 4, n_2 = 2$

Hence the first overtone frequency of the transverse vibration of the system,

$$f = \frac{n_1}{2\ell} \sqrt{\frac{F}{\mu_1}}$$

$$f = \frac{4}{2 \times 1} \sqrt{\frac{40}{0.4}} = \frac{4}{2} \times 10 = 20 \text{ Hz}$$

7. **Ans (2)**

$$E_1 = k_1 + \phi \quad \dots(i)$$

$$E_2 = k_2 + \phi \quad \dots(ii)$$

$$\text{Also, } \frac{k_1}{k_2} = 4 \Rightarrow k_1 = 4k_2 \quad \dots(iii)$$

From (i) and (ii)

$$\frac{E_1}{E_2} = \frac{k_1 + \phi}{k_2 + \phi}$$

$$\frac{\lambda_2}{\lambda_1} = \frac{4k_2 + \phi}{k_2 + \phi}$$

$$\frac{6000}{4500} = \frac{4k_2 + \phi}{k_2 + \phi}$$

$$\frac{4}{3} = \frac{4k_2 + \phi}{k_2 + \phi}$$

$$\Rightarrow 4k_2 + 4\phi = 12k_2 + 3\phi$$

$$\Rightarrow \phi = 8k_2 \quad \dots(iv)$$

Now form (ii) and (iv)

$$E_2 = k_2 + 8k_2$$

$$\Rightarrow \frac{hC}{\lambda_2} = 9k_2$$

$$\Rightarrow \frac{1242}{600} = 9k_2$$

$$\Rightarrow 2.07 = 9k_2$$

$$\Rightarrow k_2 = 0.23 \text{ eV}$$

$$\therefore \phi = 8k_2 = 8 \times 0.23 = 1.84 \text{ eV}$$

8. **Ans (1)**

$$\tau_r = -k \left(\frac{L}{2} + x \right) \theta \times \left(\frac{L}{2} + x \right)$$

$$\left(\frac{mL^2}{12} + mx^2 \right) \alpha = -k \left(\frac{L}{2} + x \right)^2 \theta$$

$$\alpha = - \frac{k \left(\frac{L}{2} + x \right)^2}{\frac{mL^2}{12} + mx^2} (\theta)$$

$$\omega^2 = \frac{k \left(\frac{L}{2} + x \right)^2}{\frac{mL^2}{12} + mx^2}$$

$$\frac{d(\omega^2)}{dx} = 0$$

$$\Rightarrow \frac{\left(\frac{mL^2}{12} + mx^2 \right) 2k \left(\frac{L}{2} + x \right) - k \left(\frac{L}{2} + x \right)^2 (2mx)}{\left(\frac{mL^2}{12} + mx^2 \right)^2} = 0$$

$$\left(\frac{mL^2}{12} + mx^2 \right) 2k \left(\frac{L}{2} + x \right) = k \left(\frac{L}{2} + x \right)^2 (2mx)$$

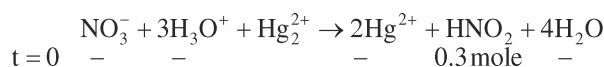
$$\frac{mL^2}{6} + 2mx^2 = mLx + 2mx^2$$

$$x = \frac{L}{6}$$

PART-2 : CHEMISTRY

SECTION-I (i)

2. **Ans (A)**



$$\text{Total charge } 0.6 F = 0.6 \times 96500 \text{ C}$$

$$Q = It = 10 \times t = 0.6 \times 96500$$

$$t = \frac{0.6 \times 96500}{10} \text{ sec.} = 1.6 \text{ hour}$$

3. **Ans (B)**

Kuchero reaction

4. **Ans (D)**

Theoretical

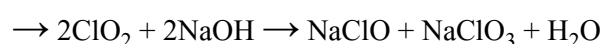
PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (A,C)**

ClO_2 is powerful oxidising agent also strong Chlorinating agent. It's bleaching power is almost 30 times stronger than Cl_2

→ In alkaline medium it undergoes disproportionation.



6. **Ans (A,B,C,D)**

(A) Ma_3b_3 type compounds shows facial and meridional isomerism.

(C) It is paramagnetic due to O_2^{-1} ligand.

(D) EDTA^{4-} containing complexes are optically active.

7. **Ans (A,B,C)**

$$\Delta S_{\text{system}} = \frac{-40.6 \times 1000}{2 \times 373} = -54.42 \text{ J/K}$$

8. **Ans (A,B,D)**

Conc. of Ag^+ required to ppt

$$\text{AgCl} = \frac{10^{-10}}{0.05} = 2 \times 10^{-9}$$

Conc. of Ag^+ required to ppt

$$\text{AgI} = \frac{4 \times 10^{-16}}{0.05} = 8 \times 10^{-15}$$

AgI give ppt first

9. **Ans (A,C,D)**

Theoretical

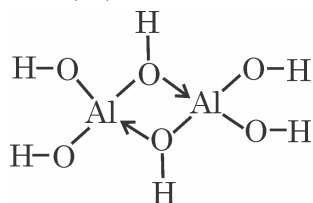
10. **Ans (B,D)**

Theoretical

PART-2 : CHEMISTRY

SECTION-III

1. **Ans (5)**



X = Total number of $(2C - 2e^-)$ bond = 10

Y = Total number of $(3C - 2e^-)$ bond = 0

Z = Total number of $(3C - 4e^-)$ bond = 2

2. **Ans (6)**

Zn, Fe, Pb, Cu, Ag, Sn produces NO_2 gas on reaction with

3. **Ans (5)**

Refer notes.

4. **Ans (2)**

e^- Jumps from 2p(since $n - 1 = 1$ and it is not s-Orbital)

to 3d(since it is associated with five possible orientations in space) and

$$\frac{5RC}{9} = RC(2)^2 \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$$

So change in $(n + \ell) = 5 - 3 = 2$

5. **Ans (6)**

$$\frac{1}{M_{\text{acid}}} = \frac{125}{1000} \times 0.1 \Rightarrow M_{\text{acid}} = 80 \text{g/mol}$$

$$0.186 = (1 + \alpha) \times 1.86 \times \left(\frac{\frac{1}{80}}{0.2} \right) \alpha = 0.6$$

6. **Ans (3)**

$$R = K(\text{NO})^a(\text{H}_2)^b$$

$$\frac{1.5}{0.25} = \left(\frac{372}{152} \right)^a \left(\frac{P}{P} \right)^b$$

$$a = 2$$

$$\frac{1.6}{0.79} = \left(\frac{P}{P} \right)^a \left(\frac{289}{144} \right)^b$$

$$b = 1$$

$$(a + b) = 3$$

7. **Ans (3)**

Theoretical

8. **Ans (7)**

Theoretical

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C)

Coefficient of x^{2r} in $(1+x+x^2)^n$

= Coefficient of x^{2r} in $(1+x(1+x))^n$

= Coefficient of x^{2r} in $\left(\left(x + \frac{1}{2} \right)^2 + \frac{3}{4} \right)^n$

Coefficient of x^{2r} in $(1+x(1+x))^n$

$$= \sum_{i=0}^r {}^nC_{i+r} {}^{r+i}C_{r-i}$$

$$= \sum_{i=0}^r \frac{{}^n}{i+r} \frac{{}^n}{n-r-i} \frac{{}^r}{2i} \frac{{}^r}{r-i}$$

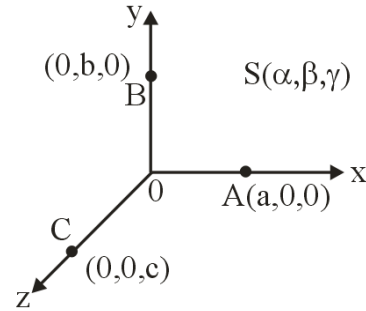
$$= \sum_{i=0}^r {}^nC_{2i} {}^{n-2i}C_{r-i}$$

Coefficient of x^{2r} in $\left(\left(x + \frac{1}{2} \right)^2 + \frac{3}{4} \right)^n$

$$= x^{2r} \text{ in } \sum_{i=0}^n {}^nC_i \left(x + \frac{1}{2} \right)^{2i} \left(\frac{3}{4} \right)^{n-i}$$

$$= \sum_{i=r}^n {}^nC_i {}^{2i}C_{2r} \left(\frac{1}{2} \right)^{2(i-r)} \left(\frac{3}{4} \right)^{n-i}$$

2. Ans (C)



$$SA \perp SB$$

$$SB \perp SC$$

$$SC \perp SA$$

$$SA \cdot SB = 0 \Rightarrow \alpha^2 + \beta^2 + \gamma^2 = a\alpha + b\beta$$

$$SB \cdot SC = 0 \Rightarrow \alpha^2 + \beta^2 + \gamma^2 = b\beta + c\gamma$$

$$SC \cdot SA = 0 \Rightarrow \alpha^2 + \beta^2 + \gamma^2 = c\gamma + a\alpha$$

$$\text{So } \frac{\alpha^2 + \beta^2 + \gamma^2}{2} = a\alpha = b\beta = c\gamma$$

$$\text{Volume} = \frac{1}{6} [\vec{a} \vec{b} \vec{c}] = 6$$

$$\Rightarrow abc = 36$$

$$\frac{(\alpha^2 + \beta^2 + \gamma^2)^3}{8\alpha\beta\gamma} = 36$$

3. Ans (D)

$$(x-1)(x^2-2)\dots\dots\dots(x^{10}-10)$$

highest power of x is 55 so we have to reduce 10

power's

10	19	127	1234
28	136		
37	145		
46	235		

$$= -10 + (9 + 16 + 21 + 24) - (14 + 18 + 20 + 30) +$$

$$(24) = 2$$

4. **Ans (B)**

$$\alpha + \beta + \gamma = -\alpha \Rightarrow 2\alpha + \beta + \gamma = 0$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \beta \Rightarrow (\alpha + \beta)\gamma - 1 = \beta$$

$$\alpha\beta\gamma = -\gamma$$

$$\Rightarrow \alpha\beta = -1$$

$$(\alpha + \beta)(-2\alpha - \beta) - 1 = \beta$$

$$\Rightarrow -2\alpha^2 - \alpha\beta - 2\beta\alpha - \beta^2 - 1 = \beta$$

$$\Rightarrow -2\alpha^2 - \beta^2 + 2 = \beta$$

$$\Rightarrow -2\alpha^2 - \frac{1}{\alpha^2} + 2 = \frac{-1}{\alpha}$$

$$\Rightarrow 2\alpha^4 - 2\alpha^2 - \alpha + 1 = 0$$

two real values of α such that exactly two solution

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,B,C,D)**

$$g'(x) = f(x)f'(x) + 25x^4 > 0$$

$$\Rightarrow f(x)f'(x) > -25x^4$$

$$\Rightarrow \int_0^1 f(x)f'(x)dx > \int_0^1 -25x^4dx$$

$$\Rightarrow \left(\frac{f^2(x)}{2}\right)_0^1 > \left(\frac{-25x^5}{5}\right)_0^1$$

$$\Rightarrow \frac{f^2(1)}{2} - \frac{f^2(0)}{2} > -5$$

$$\Rightarrow \frac{f^2(1)}{2} > 13 \Rightarrow f^2(1) > 26$$

$$\int_{-1}^0 f(x)f'(x)dx > \int_{-1}^0 (-25x^4)dx$$

$$\Rightarrow \frac{f^2(0)}{2} - \frac{f^2(-1)}{2} > -5(-1)$$

$$\Rightarrow f^2(-1) < 46$$

$$g(3) > g(0) \quad g\left(\frac{1}{2}\right) > g(0)$$

$$g(3) > 13 \quad g\left(\frac{1}{2}\right) > 13$$

6. **Ans (A,B)**

$$\lim_{n \rightarrow \infty} \frac{n^{1/3} \left[\sum_{r=1}^n \left(\frac{r}{n}\right)^{1/3} \right]}{\frac{n^{7/3}}{n^2} \sum_{r=1}^n \frac{1}{(\alpha + \frac{r}{n})^2}}$$

$$= \frac{\int_0^1 x^{1/3} dx}{\int_0^1 \frac{1}{(\alpha+x)^2} dx} = \frac{\frac{3}{4}}{\frac{-1}{\alpha+1} + \frac{1}{\alpha}}$$

$$\Rightarrow \frac{3}{4}(\alpha(\alpha+1)) = 15$$

$$\Rightarrow \alpha(\alpha+1) = 20$$

$$\alpha = 4, -5$$

7. **Ans (A,C)**

$$\alpha_n = \int_0^{1/2} \cos^n \cos nx \, dx$$

$$\alpha_{n-1} = \int_0^{1/2} \cos^{n-1} x \cos(n-1)x \, dx$$

$$\alpha_{n-1} = \int_0^{1/2} \cos^{n-1} (\cos nx \cdot \cos x + \sin nx \cdot \sin x) \, dx$$

$$\alpha_{n-1} = \int_0^{1/2} \cos^n x \cdot \cos nx \, dx + \int_0^{1/2} \underbrace{\cos^{n-1} x \sin x}_{\text{IBP}} \cdot \sin(nx) \, dx$$

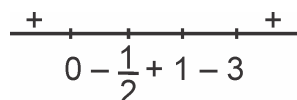
$$\alpha_{n-1} = \alpha_n + \alpha_n \Rightarrow \alpha_{n-1} = 2\alpha_n$$

$$\beta_n = \begin{cases} 0, & \text{if } n \in \text{even} \\ 1, & \text{if } n \in \text{odd} \end{cases}$$

8. Ans (B,C,D)

$f(x)$ is continuous at all $x \in \mathbb{R}$

$$f'(x) = \begin{cases} 3(x+1)^2 + 1 & , \quad x < 0 \\ 2x - 1 & , \quad 0 < x < 1 \\ x - 4 & , \quad 1 < x < 3 \\ \ln(x-2) & , \quad x > 3 \end{cases}$$



9. Ans (A,D)

$$A^1 = \begin{bmatrix} -1 & 0 & 0 \\ -1 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

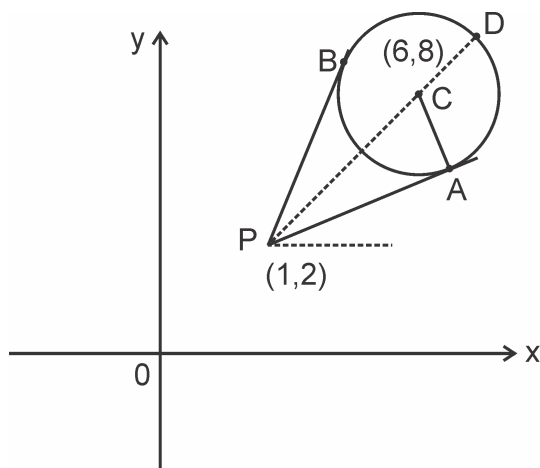
$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} -1 & 0 & 0 \\ -2 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

$$A^5 = \begin{bmatrix} -1 & 0 & 0 \\ -3 & 0 & -1 \\ -2 & -1 & 0 \end{bmatrix}$$

10. Ans (B,C,D)



eqⁿ of PA & PB

$$2 = M(x-1) \Rightarrow \frac{|5M-6|}{\sqrt{1+M^2}} = 2$$

$$= (5M-6)^2 = 4(1+M^2)$$

$$= 25M^2 - 60M + 36 = 4 + 4M^2$$

$$= 21M^2 - 60M + 32 = 0$$

$$M = M = \frac{60 \pm \sqrt{912}}{42} = \frac{30 \pm \sqrt{228}}{21}$$

least possibility to kill is when Konda is at D

$$\begin{array}{ccc}
 \sqrt{61} & & 2 \\
 \text{P} & \text{C} & \text{D} \\
 (1,2) & (6,8) & (x,y) \\
 \frac{\sqrt{61}x+2}{2+\sqrt{61}} = 6 & & \frac{\sqrt{61}y+4}{2+\sqrt{61}} = 8 \\
 x = \frac{10+6\sqrt{61}}{\sqrt{61}} & & y = \frac{12+8\sqrt{61}}{\sqrt{61}}
 \end{array}$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (5)

$$\int \frac{(t^{1/2} - 1) dt}{t(1 + t^{1/4} + t^{-1/4}) t^{1/4} \sqrt{t^{1/4} + \frac{1}{t^{1/4}}}}$$

$$t^{1/4} + \frac{1}{t^{1/4}} = v^2 \Rightarrow \left(\frac{1}{4} t^{-3/4} - \frac{1}{4} t^{-5/4} \right) dt = 2v dv$$

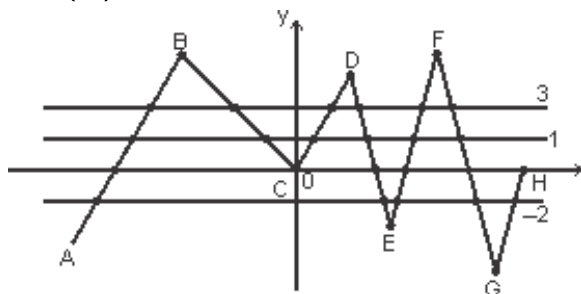
$$\Rightarrow \frac{1}{4} \left(\frac{t^{1/2} - 1}{t^{5/4}} \right) dt = 2v dv$$

$$= \int \frac{8v dv}{(v^2 + 1)v} = 8 \tan^{-1}(v) + c$$

$$= 8 \tan^{-1} \left(\sqrt{t^{1/4} + \frac{1}{t^{1/4}}} \right) + c$$

$$2g^2(16) = 2 \left(2 + \frac{1}{2} \right) = 5$$

2. Ans (7)



$$f(f(x)) = 6 \Rightarrow f(x) = -2, 1 \text{ or } 3$$

$$f(x) = -2 \text{ has 5 solutions.}$$

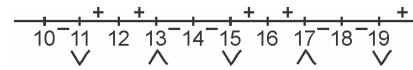
$$f(x) = 1 \text{ has 6 solutions.}$$

$$f(x) = 3 \text{ has 6 solutions.}$$

3. Ans (9)

$$g'_1(x) = \prod_{i=1}^{19} (x - i)^i$$

$$g'_1(x) = (x-1)^1 (x-2)^2 (x-3)^3 \dots (x-19)^{19}$$



$$\alpha_1 = \{19, 15, 11, 7, 3\} = 5$$

$$\beta_1 = \{17, 13, 9, 5, 1\} = 5$$

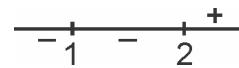
$$g'_2(x) = 98 \times 50(x-1)^{49} - 100 \times 49(x-1)^{48}$$

$$= 4900 (x-1)^{48} (x-1-1)$$

$$= 4900 (x-1)^{48} (x-2)$$

$$\alpha_2 = 1$$

$$\beta_2 = 0$$



$$g_3(x) = (x-1)^{2020} (x-2)^{2021} (x-3)^{2022}$$

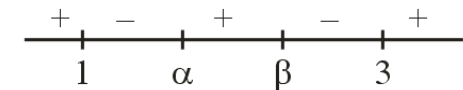
$$g'_3(x) = g_3(x) \left(\frac{2020}{x-1} + \frac{2021}{x-2} + \frac{2022}{x-3} \right)$$

$$= \frac{g_3(x)}{(x-1)(x-2)(x-3)} [2020(x-2)(x-3) + 2021(x-1)(x-3) + 2022(x-1)(x-2)]$$

$$= (x-1)^{2019} \times (x-2)^{2020} \times (x-3)^{2021} \times$$

$$[6063x^2 - 24250x + 22227]$$

$$[6063x^2 - 24250x + 22227]$$



$$\alpha_3 = 2$$

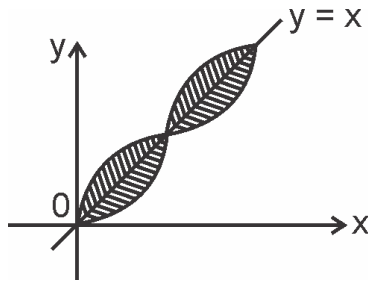
$$\beta_3 = 2$$

$$\therefore \lambda - 20 = \alpha_1 \beta_1 + \alpha_2 \beta_2 + \alpha_3 \beta_3 - 20$$

$$= 25 + 0 + 4 - 20 = 9$$

4. **Ans (4)**

$f(x)$ is symmetric about $y = x$



$$\begin{aligned}
 \text{Required area} &= 2 \int_0^{3\pi} |f(x) - x| dx \\
 &= 2 \times 3 \int_0^{\pi} (f(x) - x) dx = 2 \times 3 \int_0^{\pi} \sin x dx = 12
 \end{aligned}$$

5. **Ans (7)**

$$\begin{aligned}
 1 \ 2 \ 3 \quad & \frac{6!}{1!2!3!} \times 3! \times 2! \times 3! = 6! \times 6 \\
 2 \ 2 \ 2 \quad & \frac{6!}{2!2!2!} \times \frac{2!2!2!}{3!} \times 3! = 6!
 \end{aligned}$$

$$\text{Sum} = 6! \times (6 + 1) = 6! \times 7$$

$$\therefore \lambda = 7$$

6. **Ans (2)**

$$x(y^3 - x)dy = y(x + y^3)dx$$

$$xy^3 dy - x^2 dy = xy dx + y^4 dx$$

$$y^3 (x dy - y dx) = x^2 dy + xy dx$$

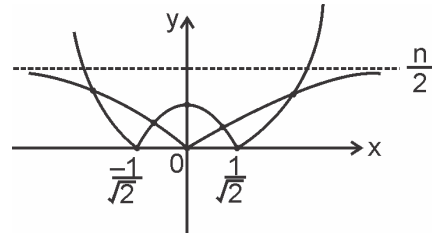
$$\Rightarrow y^3 x^2 d\left(\frac{y}{x}\right) = x d(xy)$$

$$\Rightarrow \int \frac{d(xy)}{x^2 y^2} = \int \frac{y}{x} d\left(\frac{y}{x}\right)$$

$$\Rightarrow y^3 = -2x$$

7. **Ans (4)**

$$|\tan^{-1} |2x|| = |(2x)^2 - 1|$$



8. **Ans (1)**

$$\text{let } f^{-1}(x) = g(x)$$

$$f(g(x)) = x$$

$$\Rightarrow f'(g(x)) \cdot g'(x) = 1$$

$$g'(3) = \frac{1}{f'(g(3))}$$

$$f'(x) = 3x^2 + 6x^{-4}$$

$$t + \frac{-2}{t} + 10 = 9 \Rightarrow t^2 + t - 2 = 0$$

$$\Rightarrow t = 1 \Rightarrow x^3 = 1$$

$$x = 1$$